Julian Karlbauer M.Sc.

https://jkarlbauer.github.io | julian'karlbauer@gmail.com |LinkedIn | GitHub

Applied Researcher & Engineer. My research interests lie at the intersection of spatial computing, digital fabrication, and intelligent interactive systems, with a focus on empowering and augmenting human abilities using an applied, systems-focused, and interdisciplinary approach. Expert in XR Software Engineering, and building research prototypes for industrial scale, with experience at Siemens, Audi, and Silicon Valley startups.

EXPERIENCE

Sabbatical *Latin America, Middle East, Europe, Asia (06/2025 – present)*

Planned sabbatical dedicated to cross-cultural engagement, journalistic work, independent projects, and refinement of long-term research objectives.

Phygtl Inc. *Founding Engineer* *Palo Alto, USA (remote) (12/2024 – 06/2025)*

Equity-based founding role in an XR startup dedicated to bridging physical and digital spaces through collaborative social AR experiences and enabling Generative AI–based 3D co-creation from real-world visual input. As Founding Engineer, responsible for strategic software architecture design, close collaboration with the executive team (CEO, CTO) to define the technical roadmap, leading code quality assurance, and making substantial contributions to the core codebase.

Ulm University *Research Associate* *Ulm, Germany (01/2024 – 10/2024)*

Served as a Research Associate under Prof. Dr. Enrico Rukzio in the area of Personal Fabrication. My research investigated Generative AI–assisted 3D authoring in Augmented Reality to support digital manufacturing workflows. Presented *MixGen* at ACM SCF ’24, an operational AR prototype (Python, C#) enabling generative 3D content creation from multimodal real-world input. Additionally, participated in the ACM SCF Summer School and co-authored *VRCReatIn*, a tablet-based VR 3D authoring system published at ACM MUM ’24 [3].

Siemens AG *Master’s Thesis Student* *Munich, Germany (04/2023 – 11/2023)*

Completed my Master’s thesis on data-driven quality assurance in additive manufacturing using Digital Twins and Augmented Reality interfaces. Designed, developed, and evaluated an AR system (C#, Python, JavaScript) for an industrial WAAM metal 3D printer, enabling interactive in-situ visualization of CAD data and spatio-temporal anomaly predictions. The prototype received strong institutional validation at Siemens and was selected as a potential showcase for executive leadership and presentation at Formnext.

Audi AG *Software Engineering Intern* *Ingolstadt, Germany (03/2022 – 09/2022)*

Contributed to the HCI team at the Audi Production Lab in developing a mobile, cross-device Augmented Reality platform for internal communication, leveraging 3D Digital Twin data. Led iOS-oriented Unity3D development and interaction design, and supported the design and execution of interviews and user studies.

TriCAT GmbH *R&D Intern* *Ulm, Germany (03/2021 – 06/2021)*

Proposed and developed a machine vision research prototype for in-situ, real-time object recognition and 3D localization on HoloLens 2. Implemented manually fine-tuned CNNs (Python, TensorFlow) and leveraged environment meshes within a compute-efficient remote GPU architecture. Pioneered the application of deep learning–based tools for problem-solving within the company.

Ulm University *Research Assistant* *Ulm, Germany (04/2018 – 03/2021)*

Conducted research published at ACM CHI and in ACM ToCHI, with a focus on 3D authoring [1] and perceptual aspects of Virtual Reality [2]. My responsibilities included XR prototyping, contributing to conceptual research framing, teaching activities, conducting user studies, and multimedia-based documentation and presentation of research processes for publication.

EDUCATION

Ulm University *M. Sc. in Media Informatics* *Ulm, Germany (10/2020 – 03/2024)*
Thesis Title: *"Show, Don't Tell: Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality" (Thesis Grade 1.0; Overall Grade: 1.2)*

NTNU *Visiting Student in Generative AI* *Trondheim, Norway (08/2021 – 12/2021)*
Thesis Title: "*Raw Audio Music Emotion Composition using Mood Annotated Spotify Playlists"*
(Thesis Grade 1.0; Overall Grade: 1.0)

Ulm University *B. Sc. in Media Informatics* *Ulm, Germany (04/2016 – 07/2020)*
Thesis Title: *"An Exploration of Foveated Blue Light Filters in Virtual Reality"*
(Thesis Grade 1.0)

SERVICES

- Conference Reviewer CHI 2025
- Student volunteer at the ACM SCF 2024 conference.

PUBLICATIONS

[1] Tobias Drey, Nico Rixen, **Julian Karlbauer**, and Enrico Rukzio. 2024. VRCreatIn: Taking In-Situ Pen and Tablet Interaction Beyond Ideation to 3D Modeling Lighting and Texturing. In Proceedings of the International **Conference on Mobile and Ubiquitous Multimedia (MUM '24)**. Association for Computing Machinery, New York, NY, USA, 24–35. https://doi.org/10.1145/3701571.3701580

[2] Teresa Hirzle, Fabian Fischbach, **Julian Karlbauer**, Pascal Jansen, Jan Gugenheimer, Enrico Rukzio, and Andreas Bulling. 2022. Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays. **ACM Trans. Comput.-Hum. Interact.** 29, 4, Article 33 (August 2022), 80 pages. https://doi.org/10.1145/3492802

[3] Tobias Drey, Jan Gugenheimer, **Julian Karlbauer**, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In Proceedings of the 2020 **CHI Conference on Human Factors in Computing Systems (CHI '20)**. Association for Computing Machinery, New York, NY, USA, 1–14. https://doi.org/10.1145/3313831.3376628